

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester IV Examination, 2023-24

Paper: CS-403 — Programming For Problem Solving

Time: Three hours

Maximum Marks: 70

Instructions: Question 1 is compulsory, attempt any four questions from the remaining six questions.

Question 1.

- (a) Explain five major types of programming languages with examples. [5]
- (b) What is the difference between the top-down and bottom-up approaches of programming? [5]
- (c) What are the steps involved in solving a problem? Explain your answer with an example. [4]

Question 2.

- (a) Write an algorithm to find the greatest among three numbers. [4]
- (b) Design an algorithm and flowchart to convert a decimal number to a binary number. [6]
- (c) Compare and contrast compilers and interpreters. [4]

Question 3.

- (a) Describe the order of precedence with regards to operators in the C language. [5]
- (b) Explain the different data types used in the C language. [5]
- (c) Write the steps involved in converting C source code to executable code. [4]

Question 4.

- (a) Write a C program to find the factorial of a number using recursion. [6]
- (b) What will be the outcome of the following conditional statement if the value of variable `s` is 10?
`s >= 10 && s < 25 && s != 12` [4]
- (c) Write a call-by-value function to swap two numbers. [4]

Question 5.

- (a) Explain with examples pre-increment (`++i`) and post-increment (`i++`). [4]
- (b) What is the difference between formal arguments and actual arguments? [3]

(c) What are variables, and in what way are they different from constants? [4]

(d) Explain call-by-value and call-by-reference. [3]

Question 6.

(a) Write the differences between insertion sort and selection sort. [4]

(b) Write the pseudocode of the quicksort algorithm. [5]

(c) What will be the output of the following C code?

```
#include <stdio.h>
int main() {
    int a = 1, b = 2, c = 3, d = 4, e;
    e = c + d - b * a;
    printf("%d, %d\n", e, d);
    return 0;
}
```

[5]

Question 7.

(a) How does insertion sort work? Write the pseudocode of insertion sort. Explain step-by-step how to sort an array $A = [3, 8, 5, 4, 1, 9, -2]$ in descending order. [7]

(b) What is a searching algorithm? Explain the different types of search operations. [3]

(c) Explain step-by-step how to search for the element '4' in array $A = [3, 8, 5, 4, 1, 9, -2]$ using the binary search algorithm. [4]